# FINITE PROJECTIVE MODULES OVER A HENSELIAN PAIR ARE THE SAME AS OVER ITS QUOTIENT

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*Allowed standard prerequisites (editorial).*

- Lifting a finite projective module over the quotient to a finite projective module over an etale extension which is an isomorphism on the quotient. Lemma \ref{lemma-lift-projective-module}, cited at proof lines 3364-3366. Provides the etale extension from which the module over the ring is built.
- Gabber's criterion for a henselian pair: every etale ring map inducing an isomorphism on the quotient has a section. Lemma \ref{lemma-characterize-henselian-pair}, cited at proof lines 3368-3369. Produces the section along which the lifted module is base changed.
- Nakayama's lemma. Algebra, Lemma \ref{algebra-lemma-NAK}, cited at proof lines 3376-3377. Makes a lift of an isomorphism modulo the ideal surjective.
- A surjective endomorphism of a finite module that becomes invertible after tensoring with a nilpotent quotient is invertible. Algebra, Lemma \ref{algebra-lemma-fun}, cited at proof lines 3378-3379. Makes the composition of the two surjections an isomorphism, hence both of them.

*Source reference.* Original lemma 0D4A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16445–16445. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 07M5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9715–9715. Navigation pointer to a statement the

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*Source reference.* Original lemma 09XI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12531–12531. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 00DV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 512–512. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 05G8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7052–7052. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

**Lemma 0.2.** *Let $(R, I)$ be a henselian pair. The map*

$$P \longrightarrow P/IP$$

*induces a bijection between the sets of isomorphism classes of finite projective $R$-modules and finite projective $R/I$-modules. In particular, any finite projective $R/I$-module is isomorphic to $P/IP$ for some finite projective $R$-module $P$.*

*Proof.* We first prove the final statement. Let $\overline{P}$ be a finite projective $R/I$-module. We can find a finite projective module $P'$ over some $R'$ étale over $R$ with $R/I = R'/IR'$ such that $P'/IP'$ is isomorphic to $\overline{P}$, see Lemma 07M5. Then, since $(R, I)$ is a henselian pair, the étale ring map $R \to R'$ has a section $\tau : R' \to R$ (Lemma 09XI). Setting $P = P' \otimes_{R',\tau} R$ we conclude that $P/IP$ is isomorphic to $\overline{P}$. Of course, this tells us that the map in the statement of the lemma is surjective.

Injectivity. Suppose that $P_1$ and $P_2$ are finite projective $R$-modules such that $P_1/IP_1 \cong P_2/IP_2$ as $R/I$-modules. Since $P_1$ is projective, we can find an $R$-module map $u : P_1 \to P_2$ lifting the given isomorphism. Then $u$ is surjective by Nakayama's lemma (Algebra, Lemma 00DV). We similarly find a surjection $v : P_2 \to P_1$. By Algebra, Lemma 05G8 the map $v \circ u$ is an isomorphism and we conclude $u$ is an isomorphism. □